



PROJECT ATLAS

Jetson Orin Nano Super 8GB Autonomous Rover

COMMISSIONING ADDENDUM

Final navigation, mapping, safety and reboot baseline

STATUS	PRIMARY COMMISSIONING COMPLETE
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Prepared for Dhruv | Project ATLAS | 06 August 2026

1. Current system baseline

ATLAS has completed its Raspberry Pi to Jetson migration and the primary manual-drive, encoder-odometry, SLAM, Nav2, autonomous-exploration, map-save and reboot/autostart sequence.

Subsystem	Commissioned state
Compute	NVIDIA Jetson Orin Nano Super 8GB, Ubuntu 22.04, ROS 2 Humble
Drive	Four powered wheels, two steering servos, Xbox/manual/web/Foxglove/Nav2 command mux
Navigation	RPLIDAR A1, BNO08X IMU, encoder odometry, EKF, ultrasonic secondary guard
Perception	IMX708 Camera Module 3, AI annotation, pan/tilt and face tracking
Environment	BME680 outdoor environment and AMG8833 8x8 internal thermal array
Connectivity	Wi-Fi, Tailscale, cellular telemetry, Foxglove and web dashboard
Power	BMS telemetry; Jetson 25 W mode accepted with temperature/load monitoring

Safety state

Command priority is REMOTE > WEB > FOXGLOVE > NAV2. The stale-source watchdog stops and releases an inactive channel. Explore Lite is intentionally inactive after boot and must be explicitly started. The physical emergency stop remains mandatory during ground testing.

2. Verified geometry and calibration

Parameter	Value	Engineering use
Body	0.50 x 0.36 m	Rectangular Nav2 footprint
Footprint	+/-0.25 x +/-0.18 m	Global and local costmaps
Wheelbase	0.367 m	Four-wheel steering odometry
Wheel diameter	0.125 m	0.392699 m circumference
LiDAR mount	x=-0.05, z=0.18 m, yaw=pi	Measured TF and self-filter
Inflation	0.28 m; scale 15.0	Approx. 10 cm wheel-edge target clearance
Minimum obstacle range	0.05 m	Close wall/sensor acceptance

3. Motor and encoder wiring map

Motor channel	Physical wheel	Forward PWM	Counts/rev
M1	Front-right	Positive	4048.7
M2	Front-left	Negative	3300.6
M3	Back-right	Negative	4080.1
M4	Back-left	Positive	2697.8

All four channels passed ordered forward and reverse lifted-wheel tests. Ground distance calibration was then checked using measured short moves. Do not exchange motor connectors without updating the mapping and repeating the lifted-wheel acceptance test.

4. Authoritative navigation data flow

Source	Topic / transform	Consumer / purpose
Yahboom encoders	/yahboom/odom	Measured wheel-distance odometry
BNO08X IMU	IMU topics	Robot Localization EKF
EKF	/odom and odom TF	SLAM Toolbox and Nav2
RPLIDAR A1	/scan_raw -> /scan	Self-filter, SLAM and costmaps
Static TF	base_footprint -> laser_frame	x=-0.05 m, z=0.18 m, yaw=pi
SLAM Toolbox	/map and map TF	Map creation and Nav2 global costmap
Nav2	/cmd_vel_nav	Lowest priority command-mux input
Command mux	/cmd_vel	Watchdog-controlled motor command

The base driver no longer publishes the authoritative /odom transform directly. This prevents competing odometry publishers. The EKF owns the filtered odometry path.

5. Navigation and mapping behavior

Nav2 uses `atlas_ackermann_fail_stop.xml`: one `ComputePathToPose` followed by one `FollowPath`. Planning or control failure stops the action instead of accumulating repeated recovery travel.

Explore Lite goal stability

Explore Lite now holds one active frontier goal. Changes to the frontier boundary while SLAM adds cells do not preempt that goal. A new goal is chosen only after success, abort or genuine no-progress timeout. Two consecutive Nav2 aborts request the guarded tight-space recovery path.

Mapping operating sequence

Step	Operator action	Expected ATLAS state
1	Place rover with at least 30 cm initial clearance; keep E-stop accessible	Stationary; footprint check valid
2	Verify LiDAR, odometry, IMU, map and Nav2	Sensors live; no lethal footprint cells
3	Start exploration	Mux and Explore active; one frontier goal held
4	Observe safety and map	Collision prediction aborts unsafe paths
5	Stop exploration	Mux stops; zero command; map saver runs
6	Verify files	<code>atlas_latest.yaml</code> and <code>atlas_latest.pgm</code> updated

Never start mapping when the diagnostic reports `START_POSE BLOCKED`. Move the rover manually to a clear position and recreate the fresh SLAM map before retrying.

6. Final acceptance results

Acceptance item	Result
Manual Xbox drive and priority mux	PASS
Four wheel motors and encoders	PASS
Measured short ground movement	PASS
Physical return to home	PASS; about 4 cm final error
LiDAR TF, self-filter and clear-start footprint	PASS
Bounded autonomous mapping and collision abort	PASS
Automatic safety stop and map serialization	PASS
Saved map	82 x 195 cells at 0.05 m/pixel
Reboot/autostart	PASS; 15 required services active
Unexpected autonomy after boot	NONE; Explore inactive

7. Startup and shutdown checklist

Startup

Check	Required observation
Mechanical	Wheels clear, steering free, camera mount secure, LiDAR unobstructed
Power	BMS voltage and state reasonable; no hot connector or loose ground
Compute	Jetson boots; dashboard/Foxglove connect; CPU/GPU/RAM and temperature visible
Sensors	LiDAR, IMU, ultrasonic, encoders, BME680 and AMG8833 report healthy/fresh data
Control	Xbox detected; mux active; Explore inactive until requested
Motion	Begin in light mode; verify stop before higher-power operation

Shutdown

Stop navigation/exploration, verify zero velocity, stop or park the rover, save the active map if required, then use the normal Jetson shutdown command. Do not remove battery power while map saving or filesystem writes are active.

8. Service recovery guide

Symptom	Likely cause	Safe response
Remote does not drive	Mux stopped or LB enable not held	Restart remote/mux; keep Explore off; hold LB
START_POSE BLOCKED	Nearby obstacle, self-return or stale map	Do not move autonomously; reposition and reset SLAM/Nav2
No map	SLAM startup/TF timestamp issue	Stabilize TF, restart SLAM, then Nav2
Repeated frontier goals	Old Explore binary/config	Deploy/rebuild patched Explore Lite
Nav2 overshoots short goal	Recovery tree or odometry mismatch	Use fail-stop tree; verify encoder calibration
Network lost during autonomy	Wi-Fi/cellular dropout	Use physical E-stop; rely on bounded safety timer
High CPU during mapping	SLAM/Nav2 plus visualization	Keep sensors/safety active; lower visualization priority

9. Remaining non-blocking development

Work item	Purpose	Blocking current use?
Longer full-room endurance map	Collect more drift and tight-corner evidence	No
Tight-corner frontier tuning	Improve success rate in clutter from field data	No
CrowPanel wireless controller	Replace broken 11-inch tethered display	No; separate phase
Preventive calibration	Recheck after wheel, shaft, sensor or chassis changes	Maintenance

10. Maintenance acceptance after mechanical work

Any motor, wheel, shaft, steering, LiDAR or IMU change invalidates the corresponding calibration. Lift all wheels before individual motor testing. Repeat direction, encoder sign, distance, TF and clear-start footprint checks before returning to ground autonomy.

Commissioning conclusion

The primary Project ATLAS architecture is operational and safely recoverable after reboot. Manual drive, sensor fusion, mapping, Nav2, bounded exploration and map saving have been exercised. Remaining work is field refinement and the separate CrowPanel user-interface phase, not a blocker to supervised rover operation.

END OF ADDENDUM